Appendix C: IEEE 830 Template

Software Requirements Specification

For **E-shop**

Version «# matching the last line of the versions table»

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04.03.2026.

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Revision History

Name	Date	Reason for Changes	Version
Srdjan Saranovic	04.03.2026	«"Initial version" or actual reason»	V1
Srdjan Saranovic	24.03.2026.	Added UML Use Case Diagram and completed section 1	V1.1
Srdjan Saranovic	30.03.2026.	Added UML Class diagram and complete section 2 and 3	V2

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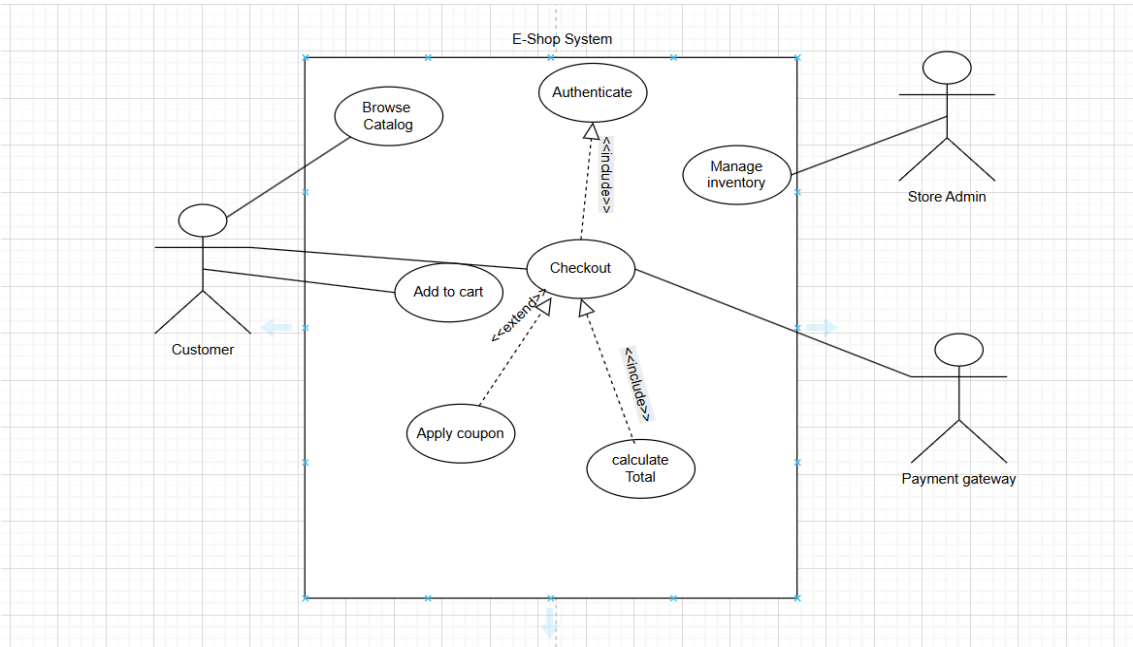
1. Introduction



1.1. Purpose

This Software Requirements Specification (SRS) defines the requirements for the E-shop System v1.0. It describes both functional and non-functional requirements for customer and admin interactions. The system enables users to browse products, add items to a shopping cart, and complete purchases through a checkout process, as illustrated in the use case diagram.

1.2. Document Conventions

This document follows standard SRS structure and uses consistent formatting:

- Functional requirements are labeled as **REQ-x.x**
- Non-functional requirements are labeled as **NF-x.x**
- Priorities are indicated (e.g., High, Medium)
- Use case names (e.g., *Checkout*, *Add to Cart*) match those shown in the diagram
  All requirements are written clearly and each has its own priority where applicable.

1.3. Intended Audience and Reading Suggestions

This document is intended for:

- **Developers** – to understand system functionality and implementation requirements
- **Testers** – to validate system behavior against requirements
- **Project managers** – to track scope and priorities
- **Designers** – to understand system interactions
- **Stakeholders** – to get an overview of system capabilities

It is recommended to first read the overview (Section 1), then the functional requirements (Section 5), followed by non-functional requirements (Section 6). The use case diagram should be reviewed alongside these sections for better understanding.

1.4. Product Scope

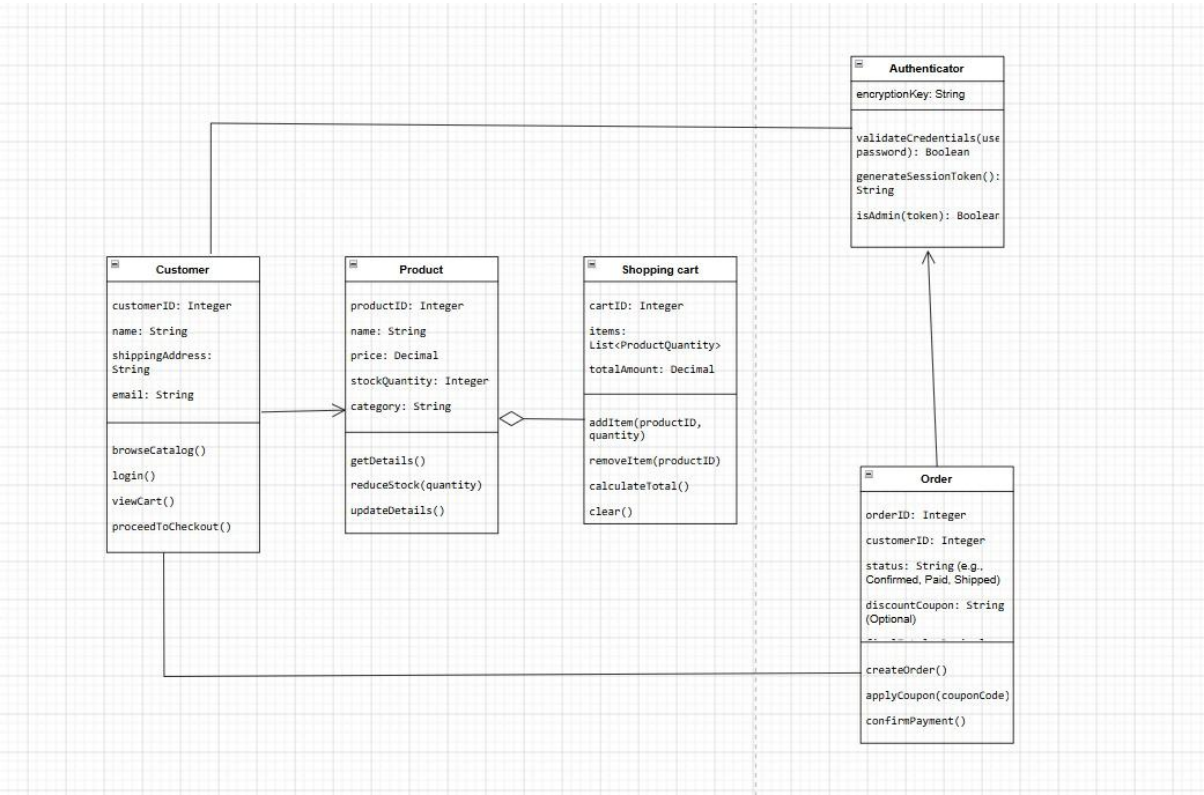
The software is an online retail platform that allows users to browse products, manage a shopping cart, and perform secure checkouts. The goal is to increase sales efficiency and provide a 24/7 shopping experience.

1.5. References

- IEEE 830 Software Requirements Specification Standard
- Course materials and assignment description
- UML Use Case Diagram guidelines
- BCrypt password hashing documentation
- SSL/TLS security standards documentation

2. Overall Description

2.1. Product Perspective



The class diagram shows the internal structure of the E-shop system. It includes main classes such as Customer, Product, Shopping Cart, Order, and Authenticator, along with their attributes and methods. The diagram also presents relationships between these classes and how they interact within the system.

2.2. Product Features

- Product Catalog Browsing.
- Shopping Cart Management.
- Secure Payment Processing.
- Order Tracking.

2.3. User Classes and Characteristics

The system has the following user classes:

- **Customer** – The main user of the system. Customers can browse products, add items to the shopping cart, and complete purchases. They have basic technical knowledge.
- **Store Admin** – Responsible for managing inventory and product information. Admin users have higher privileges and more system access.
- **Payment Gateway (External System)** – Handles payment processing. It is not a direct user but an external system interacting with the application.

The most important user class is the **Customer**, as the system is primarily designed for online shopping.

2.4. Operating Environment

The system is a web-based application that runs in a standard browser environment.

- **Platform:** Web application
- **Operating Systems:** Windows, macOS, Linux
- **Browsers:** Google Chrome, Mozilla Firefox, Safari, Edge
- **Server-side:** Application server with database support
- **Database:** Relational database (e.g., MySQL or PostgreSQL)

2.5. Design and Implementation Constraints

- The system must use **SSL/TLS encryption** for secure communication
- User passwords must be stored using **BCrypt hashing**
- The system must integrate with an external **payment gateway**
- The application should follow standard **web development practices**
- The database must support storing product, user, and order data

2.6. User Documentation

The system will include the following documentation:

- User guide for customers (how to browse, add to cart, checkout)
- Admin guide (how to manage inventory)
- Online help within the application
- Basic usage instructions on the website

2.7. Assumptions and Dependencies

- Users have internet access and a modern web browser
- The payment gateway service is available and functioning correctly
- The database system is properly configured and maintained
- Users have basic knowledge of using web applications
- The system depends on external services such as payment processing

3. System Features

This section provides a brief overview of the main system features. Detailed descriptions of these features are provided in Section 5.

3.1. Shopping Cart Management

This feature allows users to add products to the shopping cart, remove items, and update quantities. It supports the core shopping functionality of the system and is directly connected to the checkout process.

3.2. Product Catalog Browsing

This feature enables users to browse available products, view product details, and search through the catalog. It is the starting point for user interaction with the system.

3.3. Secure Checkout and Payment

This feature allows users to complete purchases by proceeding to checkout, authenticating themselves, and making secure payments through a payment gateway.

3.4. Order Management and Tracking

This feature enables users to place orders and track their status (e.g., confirmed, paid, shipped). It ensures users can monitor their purchases after checkout.

3.5. Inventory Management (Admin)

This feature allows the store administrator to manage product inventory, update product details, and control stock availability.

4. External Interface Requirements

4.1. User Interfaces Overview

«Describe the high level functionality of the system from the user's perspective. Explain how the user should be able to use your system to complete all the expected features and the feedback information that will be displayed for the user (for each screen the user will see). For example, specify whether it is GUI or text based interface and how at high level it would work. Discuss at high level how user will interact with the game such as clicking buttons, typing in some selection character, etc.»

«Describe the minimum requirements for the interface. This may include some sample screen images whether for text or GUI approach, any GUI standards or product family style guides that are to be followed, screen layout constraints, for GUI standard buttons and functions (e.g., help) that must appear on every screen, keyboard shortcuts, error message display standards, and so on»

4.2. Hardware Interfaces

«Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware, and communication protocols to be used.»

4.3. Software Interfaces

«Describe the connections between this product and other specific software components (name and version), including databases, interpreters, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Identify data that will be shared across software components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.»

5. System Features/Modules

«This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product.»

5.1. Shopping Cart Module

5.1.1 Description and Priority

Allows users to add, remove, and update quantities of items. **Priority: High**.

5.1.2 Stimulus/Response Sequences

Condition: User clicks "Add to Cart".

Response: System updates the cart count and saves the item session.

5.1.3 Functional Requirements

REQ-1.1: Upon clicking "Add to Cart", the system shall verify if the item is in stock.

REQ-1.2: The system shall calculate the total price including taxes at the checkout stage.

6. Nonfunctional Requirements

6.1. Performance

NF-1.1: The search results shall be displayed within 1.5 seconds.

6.2. Security

NF-2.1: The system shall use SSL for all payment transactions.
NF-2.2: User passwords must be hashed using BCrypt before being saved in the database.